David Mukuruva

603-349-0623 | david.d.mukuruva.ii.27@dartmouth.edu | LinkedIn | GitHub | Personal Website

Education

Dartmouth College Hanover, NH
B.A., Double Major in Mathematics and Biomedical Engineering *Expected June 2027*

Thayer School of Engineering Hanover, NH
Bachelor & Master of Engineering in Biomedical Engineering *June 2028*

Coursework: Probability (Honors), Real Analysis (Honors), Linear Algebra (Honors, proof-based), Complex Analysis (graduate), Combinatorial Game Theory, Differential Equations

Research Experience

FitzLab, Thayer School of Engineering 2026 – Present
Undergraduate Researcher, Quantum Control *Hanover, NH*

- Ran a 40-run test on Dartmouth's research supercomputer and found 2 of 8 cases frozen at a do-nothing baseline that the reported averages had hidden
- Removed one filtering step and a "clean" result's spread doubled, showing the filter drove the effect rather than the control method
- Reproduced the lab's benchmark behavior but not its published figure, and built the open-source tool the group now runs its tests on (public repo)

Blaschke Ellipses – Mathematics Research (Polymath Jr → prospective senior thesis) June 2025 – Presen
Researcher – Advisors: Prof. D. Williams, Prof. Y. Zeytuncu *Hanover, NH / Remo*

- Confirmed both target problems were still unsolved by tracing every citation of the field's main survey through 2025
- Building the first odd-order ($n = 5$) examples of a construction previously known only for even cases, and testing whether a related family of curves is always an ellipse
- Presented joint work with the Polymath Jr group at the national Joint Mathematics Meetings 2026 (contributed talk), and wrote its supporting computations in Python and Sage

Goods Lab, Thayer School of Engineering Jan 2024 – Sep 2024
Research Assistant (Computational) *Hanover, NH*

- Analyzed single-cell RNA data from cancer patients in R and mapped 16 cell types to surface candidate biomarkers
- Ran cell-cell communication analysis to map how tumor and immune cell populations signal to one another
- Built reproducible clustering pipelines in R and cross-referenced cell markers against patient metadata

Selected Projects

Studium | *local-first AI study tool — TypeScript, IndexedDB, OS Keychain* 2026 – Present

- Built a study-drill module and a local-first data path (credentials in the OS keychain, data on-device); wrote the SSRF and body-cap tests that gate the network layer
- Wired the app to multiple AI providers through user-supplied keys, generating quizzes, flashcards, and concept maps from uploaded notes

Aleph-Nought | *Next.js (TypeScript), FastAPI, PostgreSQL* 2025 – Present

- Built a symbolic engine that checks free-form math answers, inside a full-stack platform I built covering six Dartmouth math and physics course syllabi
- Automated the content pipeline with Manim, FFmpeg, Whisper, and Plotly, and shipped the stack on Vercel, Render, and Cloudflare R2

Skills & Interests

Technical: Python (NumPy/SciPy), Sage, Lean, R, C; SLURM/HPC, Git
Teaching: Lead Physics Tutor & TA, Dartmouth Emerging Engineers (10 weekly labs, 40+ students)
Interests: chess, competitive archery, combinatorial game theory